

Notes: Bisetti, Li, & Yu (RFS, 2024)

The Technical Default Spread

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1 Big picture

- **Question.** What is the quantitative impact of lender control rights on firm investment, asset prices, and macroeconomic amplification? The paper studies whether technical default — a covenant breach that reallocates control rights from the borrower to the lender — changes investment risk and therefore the cost of equity.
- **Setting.** The paper combines a dynamic general equilibrium model with endogenous loan covenants and an empirical analysis of U.S. public firms using DealScan, Compustat, and CRSP. The main firm-level sample runs from 1996 to 2016.
- **Core mechanism.** Entrepreneurs protected by limited liability want to load up on risky capital. Lenders, once they obtain control in technical default, optimally choose the safer allocation. A covenant breach therefore lowers investment risk exactly for firms that are closest to trouble.
- **Main theoretical result.** Conditional on who controls the firm, entrepreneurs choose full risky investment and lenders choose the risk-free asset. This generates a direct mapping from distance-to-technical-default into systematic risk and expected returns.
- **Main quantitative macro result.** In the calibrated model, covenants make technical default countercyclical and mitigate the financial accelerator. After a negative TFP shock combined with a positive risk shock, investment rates and capital prices fall by about 33.5% less in the economy with covenants than in the no-covenant benchmark.
- **Main asset-pricing result.** In the data, a long-short strategy that buys high-strictness firms and shorts low-strictness firms earns -4.12% per year, and the spread between the highest-strictness quintile and the fourth quintile is -7.72% . High-strictness firms also have lower exposure to the expected-investment-growth factor and to macro risk.
- **Main empirical mechanism.** Strictness predicts lower future sensitivity of investment, profitability, and cash flows to aggregate shocks, and actual covenant violations are followed by lower subsequent returns in an RDD design.
- **Why the paper matters.** The paper brings creditor control rights into macro-finance and production-based asset pricing. Its message is that lenders are not passive claimants: when control shifts, they materially reshape both firm risk-taking and aggregate propagation.

2 Data and measurement (key objects)

2.1 Loan covenant data and the distance-to-technical-default measure

- **Source.** Loan contract information comes from LPC DealScan, which covers syndicated and bilateral private loans to U.S. borrowers. The authors begin the empirical sample in 1996 because earlier covenant coverage is sparse.
- **Five covenant families.** The strictness measure is built from the five most common covenant ratios: maximum debt-to-EBITDA, minimum interest coverage, minimum fixed-charge coverage, maximum leverage, and minimum current ratio. Together they account for roughly 80% of covenant ratios in DealScan.
- **Most restrictive threshold.** If a firm has multiple packages outstanding, the authors keep the most restrictive threshold for each covenant type across all packages in a quarter.

- **Strictness.** Following Murfin (2012), strictness is the ex ante probability that the firm breaches any covenant over the next quarter. This is continuous and forward looking, unlike a realized covenant-violation dummy.
- **Economic interpretation.** Because covenant breaches reallocate control rights to lenders, strictness is a continuous measure of the likelihood that lenders will control investment decisions.

2.2 Firm-level and return data

- **Accounting data.** Compustat provides firm financials, including the accounting ratios entering strictness, profitability, leverage, investment, and balance-sheet controls.
- **Stock returns.** CRSP provides monthly stock returns, with delistings handled in the standard Fama–French style. The sample excludes financials and utilities.
- **Violation data.** Realized material covenant violations come from updated Nini–Smith–Sufi data. These data are used to validate strictness as a predictor of actual technical default.
- **Distress and constraint controls.** The paper constructs Campbell–Hilscher–Szilagyi failure probability, Bharath–Shumway EDF, the Whited–Wu index, and the Hadlock–Pierce SA index.

2.3 Macro, factor, and bank data

- **Macro risk.** Aggregate real GDP growth and real PCE growth are used both as state variables and as priced macro risks in the return analysis.
- **Cross-sectional risk factors.** The paper studies both the Fama–French five-factor model and the Hou et al. (2021) q_5 model, with special attention to the expected-investment-growth factor EG.
- **Bank data for threshold cyclicity.** For the covenant-threshold regressions, lender-level financials come from Compustat Bank and are matched to DealScan lead arrangers.

2.4 Calibration and simulated objects

- **Calibration sample.** The benchmark model is calibrated at the quarterly frequency to match U.S. annual moments from 1930 to 2016.
- **Targets.** The calibration matches the equity premium, the risk-free rate, the volatility and autocorrelation of output, consumption, and investment growth, and the average probabilities of technical and actual default.
- **Cross-sectional simulation.** To compare the model to the data, the authors simulate 800 quarters of aggregate data and then simulate 5,000 firms each period to obtain cross-sectional strictness and returns.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Representative household, pricing kernel, and production

The representative household satisfies the budget constraint

$$C_t + D_t^H = W_t L_t + R_t^D D_{t-1}^H + \Psi_t + \Psi_t^B, \quad (1)$$

and has Epstein–Zin recursive utility

$$U_t = \left\{ (1 - \beta) C_t^{1-1/\psi} + \beta \left(\mathbb{E}_t [U_{t+1}^{1-\gamma}] \right)^{\frac{1-1/\psi}{1-\gamma}} \right\}^{\frac{1}{1-1/\psi}}. \quad (2)$$

The stochastic discount factor is

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-1/\psi} \left(\frac{U_{t+1}^{1-\gamma}}{\mathbb{E}_t [U_{t+1}^{1-\gamma}]} \right)^{\frac{1/\psi - \gamma}{1-\gamma}}, \quad (3)$$

with no-arbitrage condition

$$\mathbb{E}_t [M_{t+1}] R_{t+1}^D = 1. \quad (4)$$

Firm output is Cobb–Douglas:

$$Y_{i,t} = \bar{Z}_t (e^{\omega_{i,t}} K_{i,t})^\alpha L_{i,t}^{1-\alpha}, \quad (5)$$

which implies a common marginal product of capital

$$\text{MPK}_t \equiv \alpha \bar{Z}_t \left[\frac{(1-\alpha) \bar{Z}_t}{W_t} \right]^{\frac{1-\alpha}{\alpha}} = \alpha \bar{Z}_t K_t^{\alpha-1}. \quad (6)$$

Aggregate productivity is augmented by the aggregate capital stock,

$$\bar{Z}_t = Z_t K_t^{1-\alpha}. \quad (7)$$

- **Interpretation.** The household prices all risky claims through M_{t+1} . The production block is otherwise standard, except for the capital externality in $\bar{Z}_t$, which helps generate endogenous growth and more volatile pricing kernels.
- **Why this matters.** The asset-pricing content of the model comes from the interaction between this pricing kernel, endogenous leverage, and the control-right reallocation mechanism.

3.2 Idiosyncratic productivity, the loan contract, and technical default

The firm-specific productivity shock is decomposed into an early signal and a late residual shock:

$$\omega_{i,t} = \omega_{i,t}^0 + \omega_{i,t}^1. \quad (8)$$

Entrepreneur i 's balance sheet is

$$A_{i,t} = N_{i,t} + B_{i,t} = Q_t K_{i,t+1} + D_{i,t}^E, \quad (9)$$

and the return on capital is

$$R_{t+1}^K = \frac{\text{MPK}_{t+1} + Q_{t+1}(1 - \delta)}{Q_t}. \quad (10)$$

If the realized early signal falls below the covenant threshold, control moves to the lender and the loan rate can change:

$$R_{i,t+1}^B = \begin{cases} R_{i,t+1}^{B,E}, & \omega_{i,t+1}^0 - \bar{\omega}_{i,t+1}^0 \geq 0, \\ R_{i,t+1}^{B,L}, & \omega_{i,t+1}^0 - \bar{\omega}_{i,t+1}^0 < 0. \end{cases} \quad (11)$$

Actual default occurs if the residual shock is below the cutoff $\hat{\omega}_{i,t+1}^1$, determined by

$$\left[\theta_{i,t} e^{\omega_{i,t+1}^0 + \hat{\omega}_{i,t+1}^1} R_{t+1}^K + (1 - \theta_{i,t}) R_{t+1}^D \right] A_{i,t} = R_{i,t+1}^B B_{i,t}. \quad (12)$$

- **Timing.** The covenant is written on the early signal ω^0 , which is observed after the contract is signed but before the final investment choice. That is the precise stage at which control rights are reallocated.
- **Economic meaning.** Technical default is not payment default. It is an intermediate state in which the lender takes control before final project payoffs are realized.

3.3 Entrepreneur and lender investment problems

When the entrepreneur retains control, she solves

$$V^E = \max_{\theta_{i,t}^E} \mathbb{E}_t \left[M_{t+1} \int_{\hat{\omega}_{i,t+1}^1}^{\infty} \left([\theta_{i,t}^E e^{\omega_{i,t+1}^0} R_{t+1}^K + (1 - \theta_{i,t}^E) R_{t+1}^D] A_{i,t} - R_{i,t+1}^{B,E} B_{i,t} \right) dF(\omega_{i,t+1}^1) \right]. \quad (13)$$

When the lender controls the firm, it solves

$$W^L = \max_{\theta_{i,t}^L} \mathbb{E}_t \left[M_{t+1} \left(\int_{-\infty}^{\hat{\omega}_{i,t+1}^1} (1 - \zeta) [\theta_{i,t}^L e^{\omega_{i,t+1}^0} R_{t+1}^K + (1 - \theta_{i,t}^L) R_{t+1}^D] A_{i,t} dF(\omega_{i,t+1}^1) \right. \right. \quad (14)$$

$$\left. + \int_{\hat{\omega}_{i,t+1}^1}^{\infty} R_{i,t+1}^{B,L} B_{i,t} dF(\omega_{i,t+1}^1) \right). \quad (15)$$

Proposition 1. The entrepreneur in control chooses $\theta^E = 1$, the lender in control chooses $\theta^L = 0$, and therefore $R^{B,L} = R^D$.

- **Interpretation.** Limited liability makes the entrepreneur's continuation payoff convex in project risk, so she chooses the risky project. The lender's payoff is concave because it mainly values repayment, so it chooses the risk-free asset.
- **Empirical implication.** Firms closer to technical default should have lower investment risk, lower systematic exposure, and lower expected returns because they are more likely to enter the lender-control state.

3.4 Optimal debt contract and aggregation

The contract chooses loan size, promised rate, and covenant threshold to maximize borrower value subject to lender break-even:

$$\max_{R_{i,t+1}^B, B_{i,t}, \bar{\omega}_{i,t+1}^0} V(N_{i,t}, R_{i,t+1}^B, B_{i,t}, \bar{\omega}_{i,t+1}^0), \quad (16)$$

$$\text{s.t. } W(N_{i,t}, R_{i,t+1}^B, B_{i,t}, \bar{\omega}_{i,t+1}^0) = B_{i,t}, \quad (17)$$

with ex ante values

$$V = \int_{-\infty}^{\bar{\omega}_{i,t+1}^0} V^L(\cdot, \omega_{i,t+1}^0) dF(\omega_{i,t+1}^0) + \int_{\bar{\omega}_{i,t+1}^0}^{\infty} V^E(\cdot, \omega_{i,t+1}^0) dF(\omega_{i,t+1}^0), \quad (18)$$

$$W = \int_{-\infty}^{\bar{\omega}_{i,t+1}^0} W^L(\cdot, \omega_{i,t+1}^0) dF(\omega_{i,t+1}^0) + \int_{\bar{\omega}_{i,t+1}^0}^{\infty} W^E(\cdot, \omega_{i,t+1}^0) dF(\omega_{i,t+1}^0). \quad (19)$$

Proposition 2. The optimal debt-to-net-worth ratio $H_{i,t} \equiv B_{i,t}/N_{i,t}$, the borrower-controlled loan rate $R_{i,t+1}^{B,E}$, and the covenant threshold $\bar{\omega}_{i,t+1}^0$ are identical across entrepreneurs.

- **Interpretation.** The model aggregates despite heterogeneous idiosyncratic signals. That makes it possible to study business-cycle implications in a tractable representative-agent environment.
- **Economic trade-off.** The contract internalizes two agency problems at once: equity holders want too much risk, while debt holders want too little risk.

3.5 Entrepreneur net worth and the return on equity

The entrepreneur's pre-refresh wealth satisfies

$$\frac{\tilde{N}_{i,t+1}}{N_{i,t}} = \begin{cases} R_{t+1}^D, & \omega_{i,t+1}^0 < \bar{\omega}_{i,t+1}^0, \\ e^{\omega_{i,t+1}^0} R_{t+1}^K (1 + H_t) - R_{t+1}^{B,E} H_t, & \omega_{i,t+1}^0 \geq \bar{\omega}_{i,t+1}^0 \text{ and } \omega_{i,t+1}^1 \geq \hat{\omega}_{i,t+1}^1, \\ 0, & \text{otherwise.} \end{cases} \quad (20)$$

At the aggregate level,

$$N_{t+1} = N_t \lambda \left[\int_{\bar{\omega}_{t+1}^0}^{\infty} \int_{\hat{\omega}_{t+1}^1}^{\infty} \left(e^{\omega_{t+1}^0 + \omega_{t+1}^1} R_{t+1}^K (1 + H_t) - R_{t+1}^{B,E} H_t \right) dF(\omega_{t+1}^1) dF(\omega_{t+1}^0) \right. \quad (21)$$

$$\left. + \int_{-\infty}^{\bar{\omega}_{t+1}^0} R_{t+1}^D dF(\omega_{t+1}^0) \right] + \chi N_t [1 - \lambda(1 - \Pr(\text{default}))]. \quad (22)$$

- **Interpretation.** The return on entrepreneurial equity is state contingent. In technical default, the entrepreneur only earns the safe rate; outside technical default she earns the leveraged risky return if the firm survives.
- **Asset-pricing implication.** A higher distance to technical default mechanically means more time spent in the risky-investment state and therefore a higher expected equity return.

3.6 Aggregate shocks and why risk shocks matter

Aggregate productivity follows

$$\ln Z_t - \ln \bar{Z} = \rho_z (\ln Z_{t-1} - \ln \bar{Z}) + \sigma_z \varepsilon_{Z,t}, \quad (23)$$

and the standard deviations of the early and late idiosyncratic shocks follow

$$\ln \sigma_t^j - \ln \bar{\sigma}^j = \rho_s (\ln \sigma_{t-1}^j - \ln \bar{\sigma}^j) + \sigma_{sj} \varepsilon_{s,t}, \quad j \in \{0, 1\}. \quad (24)$$

The innovations are perfectly negatively correlated:

$$\begin{bmatrix} \varepsilon_{Z,t} \\ \varepsilon_{s,t} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho_{Z,s} \\ \rho_{Z,s} & 1 \end{bmatrix} \right), \quad \rho_{Z,s} < 0. \quad (25)$$

- **Interpretation.** Bad productivity states are high-risk states. This combination raises leverage, default, and technical default exactly in recessions.
- **Quantitative lesson.** Without risk shocks, the model generates too little macro variation and even gets the cyclical of technical default wrong. Risk shocks are therefore central, not decorative.

3.7 Model-implied strictness in the cross section

Outside technical default investors observe a noisy signal,

$$\hat{\eta}_{i,t} = \omega_{i,t}^0 + \eta_{i,t}, \quad (26)$$

and model-implied strictness is defined as

$$\text{Strictness}_{i,t}^M = \begin{cases} \frac{\exp(\bar{\omega}_{t+1}^0 - \hat{\eta}_{i,t+1})}{\exp(\max_{j \in I} \{\bar{\omega}_{t+1}^0 - \hat{\eta}_{j,t+1}\})}, & \omega_{i,t+1}^0 \geq \bar{\omega}_{t+1}^0, \\ 1, & \omega_{i,t+1}^0 < \bar{\omega}_{t+1}^0. \end{cases} \quad (27)$$

- **Interpretation.** Investors do not directly observe the hidden early signal unless the firm actually enters technical default. The model therefore mirrors the empirical setting: strictness is an ex ante signal of imminent control-right transfer.

3.8 Empirical return, mechanism, and threshold specifications

The covenant-violation RDD is

$$\text{ExRet}_{i,t+3} = a + b_1 \text{Violation}_{i,t} + b_2 \text{Distance}_{i,t} + b_3 \text{Violation}_{i,t} \times \text{Distance}_{i,t} + b_4' X_{i,t} + e_{i,t+1}. \quad (28)$$

The two-factor SDF used to price strictness-sorted portfolios is

$$m_{t+1} = a - b_{\text{MKT}} \text{MKT}_{t+1} - b_{\Delta X} \Delta X_{t+1}, \quad (29)$$

where ΔX is either real GDP growth or real PCE growth.

To test the operating channel, the paper estimates

$$\Upsilon_{i,t+1} = a + b_1 \text{Strictness}_{i,t} + b_2 \Delta X_t + b_3 \text{Strictness}_{i,t} \times \Delta X_t + b_4 \Upsilon_{i,t} + c' \text{Controls}_{i,t} + \kappa_i + e_{i,t+1}, \quad (30)$$

where $\Upsilon_{i,t+1}$ is next-quarter CAPX/AT, Profit, or CF.

Finally, the cyclicity of covenant thresholds is estimated from

$$\text{Threshold}_{i,j,l,h,t} = a + b \Delta X_{t-1} + c' \text{Firm/Loan/Bank Controls} + \kappa_i + \kappa_h + e_{i,j,l,h,t}. \quad (31)$$

- **Interpretation.** These three specifications line up with the paper’s empirical logic: realized covenant breaches lower returns discontinuously; macro risk is priced in strictness-sorted portfolios; and strictness lowers the sensitivity of real decisions to aggregate shocks while covenant thresholds themselves move mildly procyclically.

4 Identification and credibility

4.1 Theory-side credibility

- **Control rights are allocated at the correct point in time.** The contract is signed before the early productivity signal is realized. After that signal arrives, the covenant decides whether the borrower or the lender allocates the firm’s assets. This timing makes the control-right channel economically sharp.
- **Different payoffs generate different choices.** The entrepreneur’s convex, limited-liability payoff produces risk shifting, while the lender’s concave repayment payoff produces conservatism. The paper does not assume different risk preferences by fiat; it derives different policies from claim structure.
- **Aggregation is nontrivial but achieved.** Proposition 2 shows that leverage, promised rates, and thresholds are identical across entrepreneurs in equilibrium, allowing the model to map micro control-right transfers into macro outcomes.

4.2 Why strictness is the key empirical variable

- **Forward looking.** Strictness measures the ex ante probability of breaching a covenant, not the ex post fact of breach.
- **Continuous.** Because it is continuous, the paper can sort portfolios and study expected returns far away from the actual breach event.
- **Validated by future violations.** In Table 5, the probability of a material covenant violation in the next quarter rises from 2% in the low-strictness portfolio to 9% in the high-strictness portfolio.

4.3 Separating control rights from distress and general financial weakness

- **Default-risk controls.** The paper controls for Campbell–Hilscher–Szilagyi failure probability and Bharath–Shumway EDF in Fama–MacBeth regressions. The strictness coefficient remains strongly negative.

- **Financial-constraint controls.** The SA and WW indices do not explain away the results, and the return pattern is similar when the sample is restricted to nondistressed firms.
- **Ratios underlying the covenant.** The authors also control for the raw accounting ratios underlying strictness, reducing the concern that strictness is just a relabeling of firm health.

4.4 Ex post breach validation

- **RDD around the covenant threshold.** Using the maximum debt-to-EBITDA covenant, the paper shows a discontinuous drop in future returns when the firm barely violates the covenant.
- **Magnitude.** Depending on specification, the estimated drop in future quarterly excess returns is about 27.8 to 42.5 basis points.
- **Why this matters.** This is the cleanest reduced-form validation that realized control-right transfers are associated with lower subsequent expected returns, exactly as the model predicts.

4.5 Risk-pricing evidence for the mechanism

- **Factor exposure.** The high-strictness portfolios have especially low exposure to the Hou et al. (2021) expected-investment-growth factor, and lower exposure to macro risks based on GDP and PCE growth.
- **Real-side sensitivity.** Strictness reduces the sensitivity of investment, profitability, and cash flows to aggregate shocks. That is what should happen if lenders in control choose safer projects.
- **Threshold cyclical.** Covenant thresholds are mildly procyclical, but technical default is still countercyclical because risk shocks dominate. This strengthens the model’s macro interpretation.

4.6 Main limitation

- **No external instrument.** The empirical analysis is not driven by a separate IV or quasi-experiment that isolates exogenous covenant tightness.
- **Contract endogeneity remains a background concern.** Covenant terms are jointly chosen with borrowers and lenders, so strictness partly reflects endogenous contracting.
- **Why the evidence is still persuasive.** The paper combines a structural theory, a continuous ex ante strictness measure, distress controls, realized-violation RDD evidence, macro risk-exposure results, and real-side sensitivity tests that all point in the same direction.

5 Tables and figures

5.1 Table 1: Calibrated parameter values

Read:

- The key calibrated ingredients are high risk aversion ($\gamma = 10$), IES ($\psi = 3$), capital share ($\alpha = 0.33$), survival probability ($\bar{\lambda} = 0.96$), and a relatively small early-signal volatility ($\bar{\sigma}_0 = 0.006$) compared with residual volatility ($\bar{\sigma}_1 = 0.281$).

- The model uses a large capital-adjustment-cost parameter ($\xi = 35$) and highly persistent TFP and volatility processes ($\rho_z = 0.988$, $\rho_s = 0.985$).
- The table matters because it shows the discipline of the calibration: the technical-default mechanism is not generated by extreme covenant parameters alone, but by a standard macro-finance calibration with lender control rights layered on top.

Table 1
Calibrated parameter values

Parameters	Symbol	Value
Capital share	α	0.330
Relative-risk aversion	γ	10
IES	ψ	3
Time discount rate	β	0.993
Depreciation rate	δ	0.025
Loss in default	ζ	0.264
Transfer to entering entrepreneurs	χ	0.528
Entrepreneur survival probability	λ	0.96
Steady state standard deviation of ω^0	$\tilde{\sigma}_0$	0.006
Steady state standard deviation of ω^1	$\tilde{\sigma}_1$	0.281
Steady-state TFP shock	$\tilde{Z}$	0.118
Capital adjustment cost	ξ	35
Persistence of TFP shock	ρ_z	0.988
Vol. of TFP shock	σ_z	0.013
Persistence of σ shock	ρ_σ	0.985
Vol. of σ_0 shock	$\sigma_{\sigma 0}$	0.017
Vol. of σ_1 shock	$\sigma_{\sigma 1}$	0.002

This table lists the parameter values used to solve and simulate the model. We calibrate the model at the quarterly frequency using data moments for the U.S. economy from 1930 to 2016.

Table 2
Aggregate moments under the benchmark and no-covenant parametrizations

Moments	Data (SD)	Benchmark	No-covenant
$\sigma(\Delta y)$	3.05(0.60)	3.12	3.40
$\sigma(\Delta c)$	2.53(0.56)	2.37	2.76
$\sigma(\Delta i)$	10.3(2.36)	7.65	10.36
$corr(\Delta c, \Delta i)$	0.40(0.28)	0.52	0.18
$corr(\Delta c, \Delta y)$	0.82(0.07)	0.88	0.75
$AC1(\Delta c)$	0.56(0.17)	0.86	0.81
$AC1(\Delta i)$	0.18(0.09)	0.20	0.05
$AC1(\Delta y)$	0.49(0.15)	0.55	0.55
$E(I_t / I_t)$	0.17(0.01)	0.27	0.28
Risk-free rate $E[R_D]$	1.1(0.16)	0.88	0.66
Volatility of real interest rate $\sigma(R_D)$	0.97(0.31)	0.72	1.01
Market return $E[R_M - R_D]$	5.71(2.25)	8.65	9.63
Loan spread $E[R^{B,E} - R^f]$	2.6	1.03	1.06
Technical default probability	28	28	
Actual default probability	3.59	2.90	3.18

This table reports key moments generated under the benchmark parameters reported in Table 1. Empirical moments are computed using U.S. annual data from 1930 to 2016. Volatilities, correlations, and first-order autocorrelations are written as $\sigma(\cdot)$, $corr(\cdot, \cdot)$, and $AC1(\cdot)$, respectively. The average annual loan spread is calculated using syndicated lending data in DealScan. The annual technical default probability comes from Chodorow-Reich and Falato (2022). Loan default rate data come from Altman and Suggitt (2000). R^D is the risk-free rate, R^M is the market return (the return on entrepreneurs' aggregate net worth), and $R^{B,E}$ is the loan rate charged by the bank when the entrepreneur is in control. To obtain moments in the model, we first simulate the model 1,000 times at the quarterly frequency based on the calibrated parameters from Table 1, and then average the resultant simulated quarterly observations at the annual level.

5.2 Table 2: Aggregate moments under the benchmark and no-covenant models

Read:

- The benchmark model matches the annual technical-default probability (28%), actual default probability (2.90%), low risk-free rate (0.88%), and high equity premium (8.65%) reasonably well.
- Relative to the no-covenant economy, the benchmark features lower volatility of output, consumption, and especially investment, plus a lower actual default probability.
- The table is the first quantitative sign that covenants mitigate amplification rather than merely reshuffle payoffs across claimants.

5.3 Table 3: Cross-sectional strictness and expected returns in the data and model

Read:

- In the data, average excess returns fall sharply in the highest-strictness quintile: the low-strictness quintile earns 6.76%, while the high-strictness quintile earns only 2.64%.
- The model reproduces the same qualitative pattern: simulated excess returns fall from 9.16% in the low-strictness portfolio to 5.71% in the high-strictness portfolio.
- The table shows that the technical-default spread is not just a reduced-form empirical fact. The calibrated model can generate it with realistic magnitudes.

5.4 Figure 1: Lender control rights and the financial accelerator

Read:

Table 3
Cross-section firm characteristics and expected returns

<i>A. Data</i>					
Firm characteristics	Low	2	3	4	High
Strictness (pp)	0.13	5.10	23.36	56.41	92.75
Excess return (pp)	6.76	8.40	6.90	10.36	2.64

<i>B. Model</i>					
Firm characteristics	Low	2	3	4	High
Strictness	37.01	44.12	49.47	56.60	78.94
Excess return (pp)	9.16	9.05	8.97	8.80	5.71

This table shows strictness and excess returns in actual data (panel A) and in simulated data (panel B) across 5 strictness-sorted portfolios. The sample period for the actual data starts in January 1996, and it ends in December 2016.

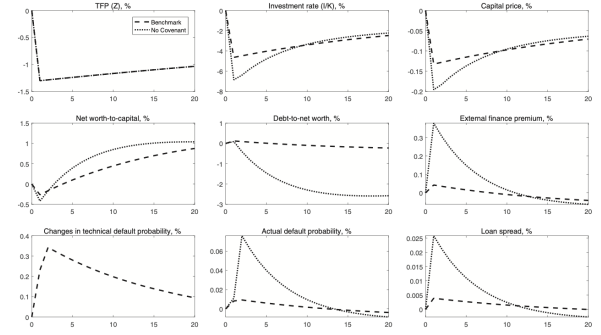


Figure 1
Lender control rights and the financial accelerator
This figure presents impulse response functions of key prices and quantities to a one-standard-deviation negative TFP shock and a one-standard-deviation perfectly correlated positive risk shock in our benchmark model ("Benchmark", the dashed black line) and in a financial accelerator model without loan covenants ("No Covenant", the dotted black line). See Table 1 for the parameters used in the model computation.

- After a negative TFP shock and a positively correlated risk shock, capital prices fall, net worth falls, leverage rises, and loan spreads rise — the standard financial-accelerator logic.
- The crucial difference is that the benchmark model generates an increase in technical default probability, which relaxes lenders' break-even conditions because they take control precisely in low-signal states and shift those firms into safe projects.
- Quantitatively, the figure shows that investment rates and capital prices decline by about 33.5% less in the covenant economy than in the no-covenant economy. This is the paper's main macro result.

5.5 Figure 2: Risk shocks and the financial accelerator

Read:

- When the authors shut down risk shocks, negative TFP shocks generate much smaller fluctuations in net worth, investment, and capital prices.
- Technical default probability actually falls in bad times in this stripped-down economy because covenant thresholds become looser while the signal distribution does not worsen.
- The figure shows why risk shocks are essential: without them, the model loses both realistic business-cycle volatility and the mitigation effect of covenants.

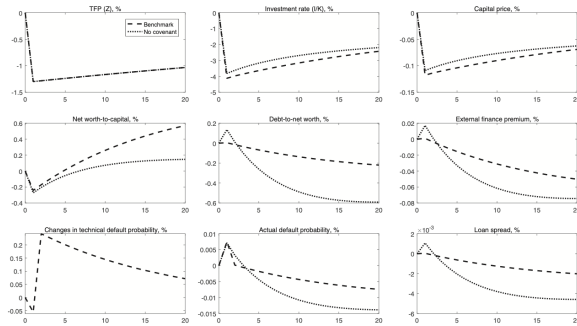


Figure 2
Risk shocks and the financial accelerator
In this figure, we study how risk shocks allow our model to generate business cycle variation and mitigation effects of the financial accelerator. The figure presents impulse response functions of key prices and quantities to a one-standard-deviation negative TFP shock in our benchmark model ("Benchmark", the dashed black line) and in a financial accelerator model without loan covenants ("No covenant", the dotted black line). We shut down risk shocks by fixing σ_T^2 and σ_T^2 at their steady-state levels. See Table 1 for the parameters used in the model computation.

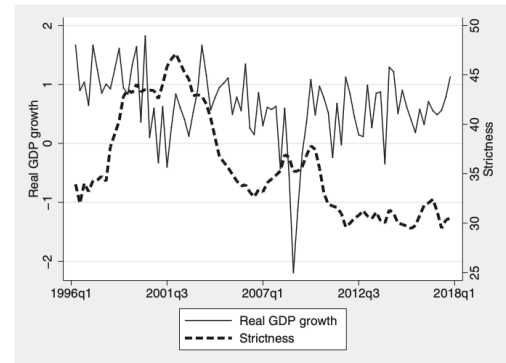


Figure 3
Covenant strictness: time series
This figure illustrates the time-series and business cycle properties of average covenant strictness in our sample (equally weighted across all firms). Strictness is the average value of firm-level strictness (defined as in Internet Appendix B.1) across all firms in a given quarter. Quarterly U.S. Real GDP growth comes from the Federal Reserve Bank of St. Louis' website. The sample starts with the first quarter of 1996 and ends with the last quarter of 2017.

5.6 Figure 3: Covenant strictness over time

Read:

- Aggregate strictness is countercyclical and tends to spike in bad times, especially around recessions.
- The series also drifts downward after the early 2000s, consistent with the longer-run decline in covenant tightness documented elsewhere.
- This figure is the reduced-form empirical counterpart to the model’s countercyclical technical-default probability.

5.7 Table 4: Summary statistics

Read:

- Strictness is economically important and very dispersed: the mean is 36.62, the median is 22.98, and the 75th percentile is 70.04.
- The firm-quarter panel has roughly 74,000 observations for strictness-related variables, while monthly return-based variables exceed 220,000 firm-month observations.
- The table also shows that realized distress measures are much smaller in median terms than strictness, reinforcing that strictness is not mechanically the same object as default probability.

5.8 Table 5: Characteristics of strictness-sorted portfolios

Read:

- High-strictness firms are more levered, less profitable, have lower credit ratings, and are closer to realized violations than low-strictness firms.
- The relation with current investment (I/K) is weak, which matters because the paper’s story is mainly about risk composition and future investment sensitivity rather than a simple cross-sectional level effect in current investment.
- The next-quarter material-violation rate rises from 2% in the low-strictness portfolio to 9% in the high-strictness portfolio, validating strictness as a forward-looking measure of lender-control likelihood.

5.9 Table 6: Portfolio returns and factor decompositions

Read:

- **Panel A.** The unconditional return spread is large and concentrated at the top of the strictness distribution: High-minus-Low earns -4.12% annually, while High-minus-4 earns -7.72% .
- **Panel B.** The Fama–French five-factor model explains little of this spread. The High-minus-Low alpha remains about -3.80% , and the High-minus-4 alpha remains about -5.77% .
- **Panel C.** The Hou et al. (2021) q_5 model absorbs much more because high-strictness firms have much lower EG exposure. The EG beta is -0.44 for High-minus-Low and -0.62 for High-minus-4, and the associated risk-premium component explains about 70.6% and 53.1% of the respective spreads.

Table 4
Summary statistics

	Mean	SD	p25	p50	p75	Observations
Strictness (pp)	36.62	36.52	2.03	22.98	70.04	74,129
Pr(Failure) (pp)	0.63	5.34	0.02	0.03	0.09	73,860
EDF (pp)	4.62	14.79	0.00	0.00	0.16	72,459
B/M	0.72	0.69	0.33	0.54	0.87	73,843
Size	6.65	1.90	5.37	6.73	7.97	221,110
I/K	0.05	0.10	0.01	0.03	0.06	60,155
ROE (pp)	2.42	8.47	0.56	2.51	4.56	74,083
Rating	1.58	1.79	0.00	0.00	3.00	74,128
Book leverage	0.29	0.17	0.16	0.27	0.40	73,755
Leverage ratio	0.29	0.18	0.16	0.27	0.40	73,756
WW index	-3.69	0.63	-4.23	-3.62	-3.25	22,191
SA index	-0.32	0.09	-0.39	-0.32	-0.26	21,371
Δ CAPX/Asset	-0.11	1.41	-0.37	-0.02	0.27	70,020
Δ ACQU/Asset	-0.21	4.69	0.00	0.00	0.00	66,911
GP (pp)	35.78	21.51	20.63	31.19	45.92	22,146
IA (pp)	13.97	36.28	-2.29	5.69	17.97	21,992
ROA (pp)	0.89	2.62	0.23	1.09	1.98	74,114
Reversal (pp)	1.05	14.10	-6.09	0.68	7.48	221,278
CF (pp)	2.18	2.52	1.31	2.24	3.28	73,445
Profit (pp)	3.48	2.30	2.24	3.32	4.59	73,478

This table presents summary statistics for the main variables in the paper. The sample starts in January 1996 and ends in December 2016, and it excludes firms in the financial and utilities industries. Strictness is constructed as in [Internet Appendix B.1](#). Failure probability (Pr(Failure)) and expected default frequency (EDF) are constructed following [Campbell, Hilscher, and Szilagyi \(2008\)](#) and [Bharath and Shumway \(2008\)](#), respectively. B/M is the book-to-market ratio (the book value of a firm's equity (SEQQ) divided by the market value of the firm's outstanding shares). Size is the natural logarithm of market capitalization. Investment rate (I/K) is investment (CAPXQ) over property, plant, and equipment (PPENTQ). Return on Equity (ROE) is net income divided by the book value of equity. Rating is a discrete score based on the firm's credit rating. Book leverage is the sum of long-term liabilities (DLTTQ) and current liabilities (DLCQ) divided by total assets (ATQ). Leverage ratio is the sum of long-term liabilities (DLTTQ) and current liabilities (DLCQ) divided by Shareholder Equity (SEQQ). Whited-Wu (WW) Index and the Size-and-Age (SA) Index are constructed as in [Whited and Wu \(2006\)](#) and [Hadlock and Pierce \(2010\)](#), respectively. Δ CAPX/Asset is the difference between four-quarters-ahead capital expenditure (CAPXQ) over property, plant, and equipment (PPENTQ). Return on Assets (ROA) is the ratio of income before extraordinary items (IBQ) over total assets (ATQ). Reversal is the 1-month-lagged return. CF is income before extraordinary items (IBQ), depreciation, and amortization (DPQ), divided by average total assets (ATQ) over the past two quarters. Profit is the ratio of operating income before depreciation (OIBDPQ) over total assets (ATQ). WW Index, SA Index, GP and IA are measured at the annual frequency. Size and Reversal are measured at the monthly frequency. All the remaining variables are measured at the quarterly frequency.

Table 5
Characteristics of strictness-based portfolios

	Low	2	3	4	High
Strictness (pp)	0.13	5.10	23.36	56.41	92.75
Excess return (pp)	6.76	8.40	6.90	10.36	2.64
Pr(Failure) (pp)	0.07	0.08	0.16	0.30	0.35
EDF (pp)	0.27	0.49	1.23	1.88	3.94
B/M	0.38	0.43	0.45	0.51	0.57
Size	10.19	10.07	9.86	9.63	9.74
I/K	0.05	0.05	0.04	0.04	0.05
ROE (pp)	6.17	4.96	4.80	3.71	1.77
Dividend payout	0.80	0.72	0.64	0.53	0.48
Rating score	4.06	3.64	3.34	2.97	2.87
Book leverage	0.20	0.27	0.32	0.35	0.42
Leverage ratio	0.85	0.98	1.32	1.69	2.20
SA index	-4.23	-4.18	-4.10	-4.05	-4.04
WW index	-0.46	-0.44	-0.42	-0.41	-0.41
Δ CAPX/Asset	-0.08	-0.03	-0.03	-0.03	-0.11
Δ ACQU/Asset	-0.18	-0.10	-0.19	-0.09	-0.34
IA (pp)	0.14	0.14	0.14	0.18	0.21
GP (pp)	0.38	0.35	0.33	0.33	0.27
Pr(Material violation) _{t+1} (pp)	0.02	0.02	0.03	0.05	0.09
Average number of firms	161.37	175.81	175.63	175.57	174.78

This table reports time-series averages of cross-sectional firm characteristics in five portfolios sorted by loan covenant strictness. The sample starts in January 1996 and ends in December 2016, and it excludes firms in the financial and utilities industries. Pr(Material Violation)_{t+1} is the portfolio-level average value of a dummy variable equal to one if a firm is in material violation of a covenant, and equal to zero otherwise. The loan covenant violation data in our sample comes from Greg Nini, and is an updated version of the data in [Nini, Smith, and Sufi \(2012\)](#). All other variables are computed as in [Table 4](#).

5.10 Table 7: Fama–MacBeth regressions of returns on strictness

Read:

- The coefficient on strictness is consistently negative, around -0.334 to -0.371 , even after controlling for size, book-to-market, reversal, leverage, ROA, failure probability, and EDF.
- This confirms that the return spread is not a disguised distress anomaly.
- The table is the paper's cleanest cross-sectional evidence that proximity to technical default lowers expected returns.

5.11 Table 8 and Figure 4: Covenant violations and future returns

Read:

- Barely breaching the most common covenant (maximum debt-to-EBITDA) is associated with about 27.8 to 42.5 basis points lower future quarterly excess returns, depending on the specification.
- The interaction term Violation $\times$ Distance is negative, showing that the return effect is steepest near the threshold.
- Figure 4 visualizes the discontinuity directly: returns step down when observations cross from the nonviolation side to the violation side of the covenant threshold.

5.12 Table 9: Portfolio-level macro risk exposures

Read:

- High-strictness firms have lower exposure to macroeconomic risk than low-strictness firms. For

Table 4
Summary statistics

	Mean	SD	p25	p50	p75	Observations
Strictness (pp)	36.62	36.52	2.03	22.98	70.04	74,129
Pr(Failure) (pp)	0.63	5.34	0.02	0.03	0.09	73,860
EDF (pp)	4.62	14.79	0.00	0.00	0.16	72,459
B/M	0.72	0.69	0.33	0.54	0.87	73,843
Size	6.65	1.90	5.37	6.73	7.97	221,110
I/K	0.05	0.10	0.01	0.03	0.06	60,155
ROE (pp)	2.42	8.47	0.56	2.51	4.56	74,083
Rating	1.58	1.79	0.00	0.00	3.00	74,128
Book leverage	0.29	0.17	0.16	0.27	0.40	73,755
Leverage ratio	0.29	0.18	0.16	0.27	0.40	73,756
WW index	-3.69	0.63	-4.23	-3.62	-3.25	22,191
SA index	-0.32	0.09	-0.39	-0.32	-0.26	21,371
Δ CAPX/Asset	-0.11	1.41	-0.37	-0.02	0.27	70,020
Δ ACQU/Asset	-0.21	4.69	0.00	0.00	0.00	66,911
GP (pp)	35.78	21.51	20.63	31.19	45.92	22,146
IA (pp)	13.97	36.28	-2.29	5.69	17.97	21,992
ROA (pp)	0.89	2.62	0.23	1.09	1.98	74,114
Reversal (pp)	1.05	14.10	-6.09	0.68	7.48	221,278
CF (pp)	2.18	2.52	1.31	2.24	3.28	73,445
Profit (pp)	3.48	2.30	2.24	3.32	4.59	73,478

This table presents summary statistics for the main variables in the paper. The sample starts in January 1996 and ends in December 2016, and it excludes firms in the financial and utilities industries. Strictness is constructed as in Internet Appendix B.1. Failure probability (Pr(Failure)) and expected default frequency (EDF) are constructed following Campbell, Hilscher, and Szilagyi (2008) and Bharath and Shumway (2008), respectively. B/M is the book-to-market ratio (the book value of a firm's equity (SEQQ) divided by the market value of the firm's outstanding shares). Size is the natural logarithm of market capitalization. Investment rate (I/K) is investment (CAPXQ) over property, plant, and equipment (PPENTQ). Return on Equity (ROE) is net income divided by the book value of equity. Rating is a discrete score based on the firm's credit rating. Book leverage is the sum of long-term liabilities (DLTTQ) and current liabilities (DLCQ) divided by total assets (ATQ). Leverage ratio is the sum of long-term liabilities (DLTTQ) and current liabilities (DLCQ) divided by Shareholder Equity (SEQQ). White-Wu (WW) Index and the Size-and-Age (SA) Index are constructed as in White and Wu (2006) and Hadlock and Pierce (2010), respectively. Δ CAPX/Asset is the difference between four-quarters-ahead capital expenditure (CAPXQ) and current capital expenditure, scaled by average assets over the same period, and Δ ACQU/Asset is the difference between four-quarters-ahead cash acquisitions (CHEQ) and current cash acquisitions, scaled by average assets over the same period. Both variables are constructed as in Nini, Smith, and Sufi (2012). GP is gross profit divided by total assets (AT), where gross profit is total revenue (REVT) minus cost of goods sold (COGS) as in Fama and French (2015). IA is the year-over-year growth rate of a firm's total assets (AT), as in Fama and French (2015). Return on Assets (ROA) is the ratio of income before extraordinary items (IBQ) over total assets (ATQ). Reversal is the 1-month-lagged return. CF is income before extraordinary items (IBQ), depreciation, and amortization (DPQ), divided by average total assets (ATQ) over the past two quarters. Profit is the ratio of operating income before depreciation (OIBDPQ) over total assets (ATQ). WW Index, SA Index, GP and IA are measured at the annual frequency. Size and Reversal are measured at the monthly frequency. All the remaining variables are measured at the quarterly frequency.

Table 8
RDD regressions: Covenant violations and future returns

	Dependent variable: <i>Quarterly returns</i>		
	(1)	(2)	(3)
Violation	-0.425*** (0.12)	-0.278** (0.12)	-0.219 (0.14)
Distance	0.137*** (0.03)	0.114*** (0.03)	0.118** (0.05)
Violation $\times$ Distance	-0.241*** (0.04)	-0.209*** (0.05)	-0.270*** (0.09)
Size		-0.023 (0.02)	-0.022 (0.02)
log B/M		0.064 (0.05)	0.065 (0.05)
Book leverage		-0.482** (0.22)	-0.477** (0.23)
ROA		4.169** (1.66)	4.139** (1.67)
High-order polynomials	No	No	Yes
Year-quarter FE	Yes	Yes	Yes
R-squared	.215	.220	.220
Observations	63,717	60,602	60,602

This table presents estimates of the RDD specification (30) using data on maximum Debt-to-EBITDA covenant violations. $Violation_{it}$ is a dummy variable equal to one if firm i 's Debt-to-EBITDA ratio exceeds its most restrictive covenant threshold in quarter t , and equal to zero otherwise. $Distance_{it}$ is the difference between firm i 's actual Debt-to-EBITDA ratio and its most restrictive covenant value. The sample starts in January 1996 and ends in December 2016. Standard errors (in parentheses) are clustered at the firm level. * $p < .1$; ** $p < .05$; *** $p < .01$.

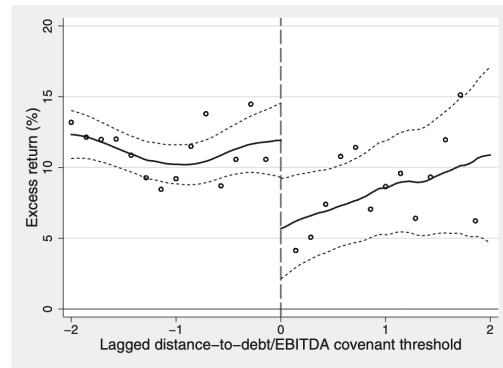


Figure 4
Covenant violations and stock returns

This figure plots average annualized excess returns against a firm's lagged distance from the Debt-to-EBITDA covenant threshold (the most common covenant in DealScan). Observations to the right of the vertical dashed line correspond to covenant violations. Each circle represents average excess returns in one of the 28 equally spaced distance bins around the threshold. The solid black lines represent fitted values of local polynomial regressions on both sides of the threshold, and the dotted black lines represent 95% confidence intervals for these estimates (using bootstrapped standard errors). The sample starts with the first quarter of 1996 and ends with the last quarter of 2017.

example, the High-minus-4 portfolio has a GDP beta of about -1.19 , and the High-minus-4 and High-minus-Low portfolios have PCE betas of about -0.86 and -0.55 .

- These results map directly into the model’s mechanism: lender-controlled firms are safer and thus load less on systematic growth risks.
- This table makes the asset-pricing story structural rather than purely statistical.

Table 9
Portfolio-level risk exposures

<i>A. Real GDP growth</i>								
	Low	2	3	4	High	High-4	High-Low	4-Low
Constant	1.73	2.20	1.75	2.80	0.76	-2.04	-0.97	1.07
<i>t</i> -stat.	5.85	7.92	4.73	6.38	1.17	-3.63	-2.22	3.89
MKT	8.67	7.54	8.55	8.44	11.53	3.09	2.85	-0.24
<i>t</i> -stat.	25.06	14.25	21.03	8.27	35.51	3.43	7.23	-0.27
Δ GDP	-0.62	-0.10	-0.95	0.16	-1.03	-1.19	-0.41	0.78
<i>t</i> -stat.	-0.97	-0.24	-1.45	0.33	-1.34	-1.98	-1.53	1.24

<i>B. Personal consumption expenditure growth</i>								
	Low	2	3	4	High	High-4	High-Low	4-Low
Constant	1.73	2.20	1.75	2.80	0.76	-2.04	-0.97	1.07
<i>t</i> -stat.	6.62	8.70	7.38	6.22	1.52	-3.85	-2.64	3.65
MKT	8.53	7.55	8.33	8.59	11.29	2.70	2.76	0.06
<i>t</i> -stat.	23.21	14.98	16.28	9.06	29.10	3.38	6.01	0.08
Δ PCE	-0.83	-0.36	-1.23	-0.53	-1.38	-0.86	-0.55	0.30
<i>t</i> -stat.	-1.68	-0.62	-2.67	-0.98	-2.16	-2.28	-2.43	1.29

This table reports factor loadings of strictness-sorted portfolios from two-factor time-series regressions of excess returns on market excess returns (MKT) and macroeconomic shocks. In panel A, we use the growth rate of quarterly real GDP (Δ GDP) in the United States as a macroeconomic shock. In panel B, we use the growth rate of quarterly real personal consumption expenditure (Δ PCE) in the United States as a macroeconomic shock. Nominal quantities and corresponding price indexes come from tables 1.1.5 and 1.1.4 of the National Income and Products Accounts (NIPA) from the U.S. Bureau of Economic Analysis (BEA). The *t*-statistics are computed based on Newey and West (1987) adjusted standard errors with 36 lags. The sample starts in January 1996 and ends in December 2016.

Table 10
Estimating the price of risk

	(1)	(2)	(3)
MKT	0.36	-0.02	0.15
<i>t</i> -stat.	3.26	-0.13	1.00
Δ GDP		1.01	
<i>t</i> -stat.		4.04	
Δ PCE			0.91
<i>t</i> -stat.			3.41
SSQE(%)	0.26	0.29	0.14
MAPE(%)	2.53	2.58	1.72
<i>J</i> -statistic	14.70	11.47	11.46
<i>p</i> -value	.14	.24	.25

This table presents the results of a two-factor SDF GMM estimation. The first factor in the SDF is the market excess return (MKT), and the second factor is either the growth rate of quarterly U.S. real GDP (Δ GDP, column 2), or the growth rate of U.S. real personal consumption expenditure (Δ PCE, column 3). We normalize all factors such that $E[M] = 1$ (Cochrane 2009) and report *t*-statistics based on Newey-West standard errors adjusted for three lags. As measures of model fit, we report the estimated sum of squared errors (SSQE), the mean absolute pricing errors (MAPE). As a measure of model overidentification, we report the *J*-statistic. We use 6 strictness-sorted portfolios (the five quintile-based portfolios and the High-Low portfolio) and the 6 expected-investment growth portfolios of Hou et al. (2021) as test assets, following Lewellen, Nagel, and Shanken (2010). The sample starts in January 1996 and ends in December 2016.

5.13 Table 10: Estimating the price of macro risk

Read:

- Real GDP growth and real PCE growth are priced risk factors in two-factor SDF estimations. The estimated prices of risk are about 1.01 for GDP growth and 0.91 for PCE growth.
- Adding PCE growth to the market factor improves pricing fit, lowering SSQE from 0.26% in the CAPM to 0.14%.
- The table supports the interpretation that lower macro-risk exposure is the reason high-strictness firms earn lower expected returns.

5.14 Table 11: Investment, profitability, and cash-flow sensitivity to aggregate risk

Read:

- Strictness lowers the sensitivity of future investment, profitability, and cash flows to aggregate shocks. The interaction terms Δ GDP $\times$ Strictness and Δ PCE $\times$ Strictness are strongly negative in most specifications.
- The paper’s economic examples are intuitive: a one-standard-deviation increase in Δ GDP raises next-quarter investment for a zero-strictness firm but slightly lowers it for a unit-strictness firm; similarly, a one-standard-deviation increase in Δ PCE raises profitability for a zero-strictness firm but has roughly no effect for a unit-strictness firm.
- This is perhaps the strongest direct evidence for the mechanism: strictness is associated with safer project choice, not just lower valuations.

Table 11
Investment, profitability, and cash flow sensitivity to aggregate risk

	CAPX/AT _{t+1}		Profit _{t+1}		CF _{t+1}	
	(1)	(2)	(3)	(4)	(5)	(6)
Strictness	-0.080*** (0.03)	-0.058* (0.03)	-0.246*** (0.05)	-0.244*** (0.06)	-0.154** (0.06)	-0.199** (0.08)
Δ GDP	5.217*** (0.96)		11.995*** (1.67)		21.801*** (2.45)	
Δ GDP × Strictness	-6.173*** (1.89)		-17.142*** (3.22)		-13.331*** (5.05)	
Δ PCE		14.461*** (2.03)		18.870*** (3.22)		35.858*** (4.57)
Δ PCE × Strictness		-10.467*** (3.83)		-18.005*** (6.09)		-4.444 (9.72)
Size	-0.063*** (0.01)	-0.060*** (0.01)	0.197*** (0.02)	0.200*** (0.02)	0.390*** (0.03)	0.397*** (0.03)
Book leverage	-1.117*** (0.10)	-1.130*** (0.10)	-0.000 (0.14)	-0.017 (0.14)	-0.752*** (0.15)	-0.800*** (0.15)
log B/M	-0.288*** (0.02)	-0.281*** (0.02)	-0.679*** (0.03)	-0.674*** (0.03)	-0.774*** (0.03)	-0.766*** (0.03)
Profit	0.034*** (0.00)	0.034*** (0.00)	0.302*** (0.02)	0.301*** (0.02)	0.220*** (0.01)	0.216*** (0.01)
CAPX/AT	0.399*** (0.01)	0.398*** (0.01)				
CF					0.015 (0.01)	0.018* (0.01)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.707	.707	.559	.559	.411	.411
Observations	71,979	71,979	67,313	67,313	67,278	67,278

In this table, we use panel regressions to study the interaction between covenant strictness and the sensitivity of future investment, profitability, and cash flows to aggregate risk. In column 1, we regress one-quarter-ahead capital-expenditure-to-assets ratio (CAPX/AT) on strictness, ΔGDP, and their interactions. In column 2, we replace ΔGDP (and its interaction with strictness) with ΔPCE. In columns 3 and 4, we use one-quarter-ahead profitability (Profit) as the dependent variable, and in columns 5 and 6 we use one-quarter-ahead cash flows to assets (CF) as the dependent variable. The sample starts in January 1996 and ends in December 2016. Standard errors (in parentheses) are clustered at the firm level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 12
Cyclical pattern of covenant thresholds

	Dependent variable: <i>Covenant thresholds</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Lagged Δ GDP	0.495** (0.24)	0.590** (0.29)	0.717** (0.29)	0.707** (0.30)				
Lagged Δ PCE					0.663* (0.40)	0.895* (0.48)	1.105** (0.48)	1.411*** (0.53)
Firm-level controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Loan-level controls	No	No	Yes	Yes	No	No	Yes	Yes
Lender-level controls	No	No	No	Yes	No	No	No	Yes
R-squared	.380	.423	.427	.425	.380	.423	.427	.425
Observations	42,343	27,680	27,564	21,796	42,343	27,680	27,564	21,796

In this table, we estimate panel regressions to study the cyclical pattern of covenant thresholds. We first build a quarterly unbalanced panel at the borrower-lender-package-covenant type level containing, for each newly issued loan on DealScan, information about the borrower, the lender, and the loan. Loan information includes maturity, amount, and threshold values for the five covenant types we use to compute strictness. For covenants on maximum values (e.g., limits on the maximum debt-to-EBITDA ratio) we use reciprocals, and to ease coefficient interpretation we normalize the five thresholds by their corresponding average in each four digit-SIC industry. We consider two real business cycle indicators as independent variables, namely, the quarterly growth rate of U.S. real GDP (ΔGDP), and the quarterly growth rate of U.S. real personal consumption expenditure (ΔPCE). Both ΔGDP and ΔPCE are lagged by one quarter. In column 1, we regress covenant thresholds at the firm-loan-covenant-lender-quarter level on one-quarter-lagged ΔGDP, firm fixed effects, and lender fixed effects. In columns 2 and 3, we incrementally include firm-level controls (the five ratios underlying the covenants) and loan-level controls (loan amount and maturity) to the specification. In column 4, we also include lender-level controls, namely, the Tier 1 ratio and leverage (total liabilities to total assets). This bank-level information comes from Compustat bank. If the same package has more than one lead arranger, we assign repeated observations of the dependent variables to different lead arrangers. In columns 5 to 8, we repeat the same estimation exercise while replacing lagged ΔGDP with lagged ΔPCE as the independent variable. The sample starts in January 1996 and ends in December 2016. Standard errors (in parentheses) are double clustered at the firm-bank level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.15 Table 12: Cyclical pattern of covenant thresholds

Read:

- Covenant thresholds are mildly procyclical. Depending on controls, a one-percentage-point increase in lagged real GDP growth is associated with about a 0.50 to 0.71 increase in normalized threshold tightness; the corresponding PCE coefficients run from about 0.66 to 1.41.
- The direction matters: covenants get slightly looser in bad times, which by itself would lower technical default probability.
- The fact that technical default remains countercyclical despite this procyclicality shows how important the worsening signal distribution is in recessions.

6 Short summary

- **Conclusion.** Covenant-induced lender control rights matter for both macro amplification and the cross section of expected stock returns.
- **Intuition.** Entrepreneurs want risk because they have limited liability; lenders want safety because they care about repayment. Technical default hands the wheel to the safe player.
- **Empirical lesson.** Distance to technical default prices equity because it forecasts who will choose low-risk investment policies next.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Technical default reallocates control rights in a way that lowers firm risk and lowers expected equity returns.

- **Macro implication.** Because technical default is countercyclical, covenants soften the financial accelerator instead of worsening it.
- **Cross-sectional implication.** Firms closest to technical default earn lower expected returns because they are less exposed to aggregate investment-growth and macroeconomic risk.
- **Validation.** The same mechanism appears ex ante in strictness sorts and ex post around realized covenant violations.
- **Quantitative importance.** The model matches a sizable technical-default spread, and the empirical return spread is economically large at -4.12% annually for High-minus-Low.

7.2 Economic intuition

- Debt contracts do more than set prices; they determine who controls capital allocation after bad signals arrive.
- The same covenant that protects lenders also changes the firm's systematic risk by changing who chooses the project mix.
- As a result, lender control rights belong in macro-finance and production-based asset pricing, not only in corporate-contracting studies.

7.3 Takeaway

This paper makes lender control rights quantitatively central. In the model, covenant breaches hand investment decisions to lenders, who choose safer allocations and soften macro amplification. In the data, firms near technical default load less on systematic risk and earn lower average returns, linking creditor rights, business-cycle propagation, and the cross section of stock returns in one framework.